Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	03 october 2026
Team ID	**SWTID-2026-7991**
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Technical Architecture:

The architecture below shows the ServiceNow user, ACL configuration, application table and access-decision flow.

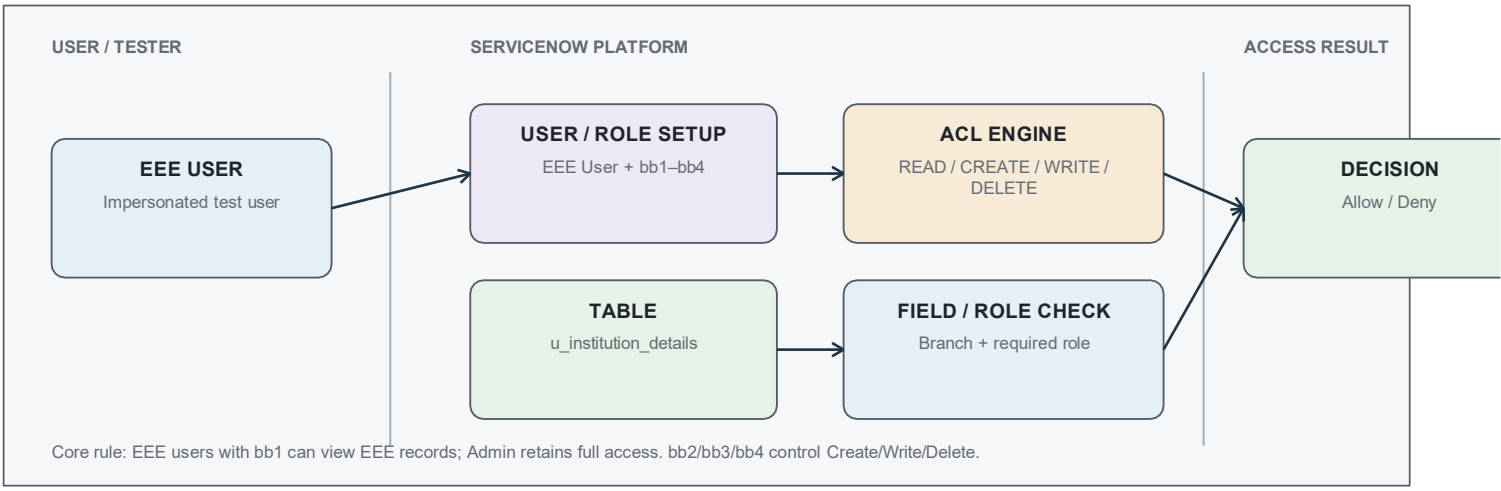


Table-1 : Components & Technologies:

Architecture Guidelines

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the application and performs impersonation/testing.	ServiceNow web interface
2	Application Logic-1	Access-control logic for record READ based on role and Branch value.	ServiceNow ACL + JavaScript
3	Application Logic-2	Access-control logic for CREATE and required role evaluation.	ServiceNow ACL framework
4	Application Logic-3	Access-control logic for WRITE and DELETE operations.	ServiceNow ACL framework
5	Database	Stores Institution Details records and fields such as Branch, Email and Phone.	ServiceNow platform database
6	Cloud Database	Separate cloud database is not specified in the supplied lab.	Not specified / N.A.
7	File Storage	No file-storage component is required by the supplied ACL lab.	Not specified / N.A.
8	External API-1	No external API is required for the ACL configuration.	Not specified / N.A.
9	External API-2	No external API is required for the ACL configuration.	Not specified / N.A.
10	Machine Learning Model	No machine-learning model is used in this lab.	Not applicable
11	Infrastructure (Server / Cloud)	Application and ACL configuration are executed in the ServiceNow instance.	ServiceNow platform / instance

Source content: Script-Controlled ACL – Restrict Record Access Based on Field Value. Layout modeled on the uploaded Technology Stack template.

Project Design Phase-II
Technology Stack – Application Characteristics

Table-2: Application Characteristics:

Application characteristics for the Script-Controlled ACL implementation.

S.No	Characteristics	Description	Technology
1	**Open-Source Frameworks**	The supplied lab does not specify a separate open-source framework; the implementation is performed within ServiceNow.	ServiceNow platform
2	**Security Implementations**	Record-level ACLs restrict access using roles and the Branch field. READ uses bb1 and Branch = EEE; Admin retains full access. CREATE/WRITE/DELETE use bb2/bb3/bb4.	ServiceNow ACLs, roles, JavaScript
3	**Scalable Architecture**	Permissions are separated by operation-specific ACLs and custom roles, allowing additional roles, users, records or rules to be added.	ServiceNow ACL framework
4	**Availability**	The lab is executed in an active ServiceNow instance; the required table and ACL records must remain active for access decisions to work.	ServiceNow instance
5	**Performance**	The ACL checks are applied as part of normal record access operations. The supplied lab does not specify request-per-second targets, caching or CDN requirements.	ServiceNow ACL evaluation

ACL Permission Summary

Role	Operation	Required condition	Expected access
bb1	READ	Branch is EEE	View EEE branch records
bb2	CREATE	Role check; no data condition	Create records
bb3	WRITE	Role check; no data condition	Edit permitted records
bb4	DELETE	Role check; no data condition	Delete permitted records
admin	ALL	Admin bypass in supplied READ script	Full READ access regardless of branch

Verification Flow

Project Technology Summary

The project is a ServiceNow-based record-security lab. Its core technology is the ServiceNow ACL framework, with JavaScript used in the Advanced READ ACL script. The custom u_institution_details table provides the protected records, while bb1–bb4 roles control READ, CREATE, WRITE and DELETE behavior. The supplied lab does not define external APIs, machine-learning models, separate cloud databases, or file storage.

Reference model: uploaded Technology Stack (Architecture & Stack) template, including Table-1 and Table-2 structure.

Project content: Script-Controlled ACL lab.